Chapter 6. Comparing Mean Vectors and MANOVA

Paired comparisons, contrasts, two populations, one- and two-way MANOVA, simultaneous intervals

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1 Paired Comparisons and Repeated Measures

1.1 Paired comparisons

[Supplement] When two treatments are applied to the same unit and p responses are measured, analyze the differences $\mathbf{D}_j = \mathbf{X}_{1j} - \mathbf{X}_{2j}$, $j = 1, \ldots, n$; unit effects cancel, increasing precision.

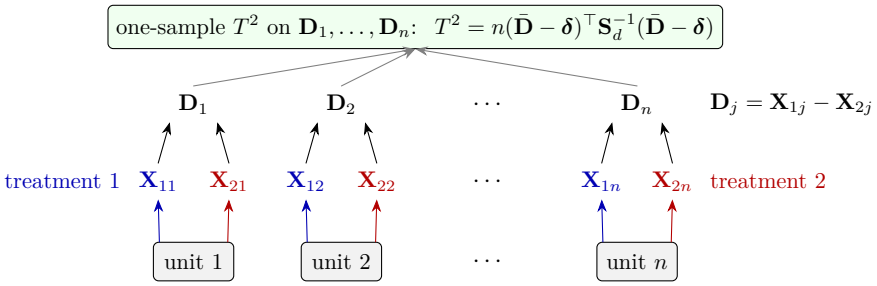


Figure. *[Supplement] Paired design: differences within a unit turn the comparison into a one-sample problem for $\boldsymbol{\delta} = \mathrm{E}\mathbf{D}_j$.*

Theorem 6.1 (Result 6.1): paired T^2

Let $\mathbf{D}_1, \ldots, \mathbf{D}_n$ be a random sample from $N_p(\boldsymbol{\delta}, \boldsymbol{\Sigma}_d)$. Then

$$T^2 = n(\bar{\mathbf{D}} - \boldsymbol{\delta})^\top \mathbf{S}_d^{-1}(\bar{\mathbf{D}} - \boldsymbol{\delta})$$

is distributed as $\frac{(n-1)p}{n-p} F_{p,n-p}$, whatever the true $\boldsymbol{\delta}$ and $\boldsymbol{\Sigma}_d$. If n and $n-p$ are both large, T^2 is approximately χ^2_p regardless of the population of differences.

Proof. [Supplement] This is the one-sample T^2 (Theorem 5.3) applied to $\mathbf{D}_j$: $(n-1)\mathbf{S}_d \sim W_p(n-1, \boldsymbol{\Sigma}_d)$, independent of $\bar{\mathbf{D}}$ (the slide prints $(n-p)\mathbf{S}_d$; the degrees of freedom are $n-1$). Without normality use Theorem 4.12. ■

[Supplement] The test of $H_0 : \boldsymbol{\delta} = \mathbf{0}$, the confidence region, simultaneous intervals $\bar{d}_i \pm \sqrt{\frac{(n-1)p}{n-p} F_{p,n-p}(\alpha)} \sqrt{s^2_{d_i}/n}$ and Bonferroni intervals $\bar{d}_i \pm t_{n-1}(\alpha/2p)\sqrt{s^2_{d_i}/n}$ are as in Ch. 5.

1.2 Contrasts and repeated measures designs

[Supplement] With q repeated measurements $\mathbf{X}_j \in \mathbb{R}^q$ on each unit, "all treatment means are equal" is a hypothesis $\mathbf{C}\boldsymbol{\mu} = \mathbf{0}$, e.g. $\mathbf{C} = \begin{pmatrix} -1 & 1 & 0 & \cdots \\ -1 & 0 & 1 & \cdots \\ & & \ddots & \end{pmatrix}$ ($(q-1) \times q$, rows are contrasts).

Theorem 6.2: test for contrasts

If $\mathbf{X}_j \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{C} \in \mathbb{R}^{q \times p}$ has rank q, then under $H_0 : \mathbf{C}\boldsymbol{\mu} = \mathbf{b}$

$$T^2_c = n(\mathbf{C}\bar{\mathbf{x}} - \mathbf{b})^\top (\mathbf{C}\mathbf{S}\mathbf{C}^\top)^{-1}(\mathbf{C}\bar{\mathbf{x}} - \mathbf{b}) \sim \frac{(n-1)q}{n-q} F_{q,n-q} = \frac{q(n-1)}{(n-1)-(q-1)} F_{q,n-q}.$$

Proof. (i) $\sqrt{n}(\mathbf{C}\bar{\mathbf{X}} - \mathbf{C}\boldsymbol{\mu}) \sim N_q(\mathbf{0}, \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^\top)$;

(ii) $(n-1)\mathbf{C}\mathbf{S}\mathbf{C}^\top \sim W_q(n-1, \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^\top)$ (Theorem 4.2(i)), independent of $\mathbf{C}\bar{\mathbf{X}}$ since $\bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}$. Apply Theorem 5.2 in dimension q with $m = n-1$. ■

[Supplement] By invariance (Theorem 5.4), T_c^2 does not depend on which contrast matrix with the same row space is used.

2 Comparing Mean Vectors from Two Populations

Motivation. Two-group comparisons generalize the two-sample t-test. With a common covariance matrix, the pooled $\mathbf{S}$ gives an exact T^2 test; with unequal covariances, large-sample approximations are used.

Remark 6.3: comparing two distributions versus two means

The general question is $H_0 : F_1 = F_2$ versus $H_1 : F_1 \neq F_2$ for the two population distributions. If both populations are normal with a common covariance, $F_\ell = N_p(\boldsymbol{\mu}_\ell, \boldsymbol{\Sigma})$, then $F_1 = F_2 \iff \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$, so the problem reduces to $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$ versus $H_1 : \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$. Without these assumptions, equal means does not imply equal distributions.

Remark 6.4: the univariate case $p = 1$

With samples $X_1, \ldots, X_m$ and $Y_1, \ldots, Y_n$, the pooled two-sample t statistic is

$$T = \left(\frac{1}{m} + \frac{1}{n}\right)^{-1/2} \frac{\bar{X}_m - \bar{Y}_n}{S_p}, \qquad S_p^2 = \frac{\sum_{i=1}^m (X_i - \bar{X}_m)^2 + \sum_{i=1}^n (Y_i - \bar{Y}_n)^2}{m + n - 2},$$

and $T \sim t_{m+n-2}$ under H_0. For $p \geq 1$ the pooled covariance $\mathbf{S}_p = (s_{ij})$ has entries

$$s_{ij} = \frac{\sum_{k=1}^m (X_{ki} - \bar{X}_{\cdot i})(X_{kj} - \bar{X}_{\cdot j}) + \sum_{k=1}^n (Y_{ki} - \bar{Y}_{\cdot i})(Y_{kj} - \bar{Y}_{\cdot j})}{m + n - 2},$$

and the multivariate statistic is the squared norm generalizing T^2:

$$T^2 = \left(\frac{1}{m} + \frac{1}{n}\right)^{-1} \|\mathbf{S}_p^{-1/2}(\bar{\mathbf{X}}_m - \bar{\mathbf{Y}}_n)\|_2^2 \sim \frac{(m + n - 2)p}{m + n - p - 1} F_{p,m+n-p-1}.$$

For $p = 1$ this is exactly the square of the t statistic, since $t_\nu^2 = F_{1,\nu}$. The p-value is $P(T^2 > T^2_{\text{obs}} \mid H_0) = P(F_{p,m+n-p-1} \geq F_{\text{obs}})$ with $F_{\text{obs}} = \frac{m+n-p-1}{(m+n-2)p} T^2_{\text{obs}}$. This follows from the general form $T^2 = \mathbf{z}^\top (\mathbf{W}/\nu)^{-1} \mathbf{z} \sim \frac{p\nu}{\nu-(p-1)} F_{p,\nu-(p-1)}$ for $\mathbf{z} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$ independent of $\mathbf{W} \sim W_p(\nu, \boldsymbol{\Sigma})$ (Theorem 5.2 with $m = \nu$), here with $\nu = m + n - 2$.

2.1 Common covariance: the pooled T^2

$\mathbf{X}_{11}, \ldots, \mathbf{X}_{1n_1} \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$ and an independent sample $\mathbf{X}_{21}, \ldots, \mathbf{X}_{2n_2} \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$. Pooled covariance:

$$\mathbf{S}_{\text{pooled}} = \frac{(n_1 - 1)\mathbf{S}_1 + (n_2 - 1)\mathbf{S}_2}{n_1 + n_2 - 2}.$$

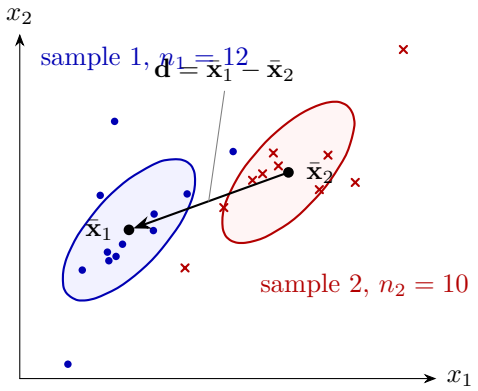


Figure. *Two samples with a common* $\boldsymbol{\Sigma}$*: the same contour of* $\mathbf{S}_{pooled}$ *about* $\bar{\mathbf{x}}_1$ *and* $\bar{\mathbf{x}}_2$ *(here* $T^2 = 57.1$*).*

Theorem 6.5 (Result 6.2): two-sample T^2

$$T^2 = (\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2))^\top \left[\left(\tfrac{1}{n_1} + \tfrac{1}{n_2}\right)\mathbf{S}_{\text{pooled}}\right]^{-1} (\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2))$$
$$\sim \frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - p - 1} F_{p, n_1+n_2-p-1}.$$

Consequently $P(T^2 \le c_\alpha^2) = 1 - \alpha$ with $c_\alpha^2 = \frac{(n_1+n_2-2)p}{n_1+n_2-p-1} F_{p,n_1+n_2-p-1}(\alpha)$.

Proof. (i) $\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 \sim N_p(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2, (\frac{1}{n_1} + \frac{1}{n_2})\boldsymbol{\Sigma})$, so $\mathbf{z} = (\frac{1}{n_1} + \frac{1}{n_2})^{-1/2}\{\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)\} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$.
(ii) $(n_i - 1)\mathbf{S}_i \sim W_p(n_i - 1, \boldsymbol{\Sigma})$ independently, so $(n_1 + n_2 - 2)\mathbf{S}_{\text{pooled}} \sim W_p(n_1 + n_2 - 2, \boldsymbol{\Sigma})$ by Theorem 4.2(iii).

(iii) $\bar{\mathbf{X}}_i \perp\!\!\!\perp \mathbf{S}_i$ within each sample and the samples are independent, so $\mathbf{z} \perp\!\!\!\perp \mathbf{S}_{\text{pooled}}$. Hence $T^2 = \mathbf{z}^\top \mathbf{S}_{\text{pooled}}^{-1} \mathbf{z} = (\frac{1}{n_1} + \frac{1}{n_2})^{-1} \|\mathbf{S}_p^{-1/2}(\bar{\mathbf{x}} - \bar{\mathbf{y}} - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2))\|_2^2 \sim T_p^2(n_1 + n_2 - 2)$, and Theorem 5.2 with $m = n_1 + n_2 - 2$ gives the F form. ∎

The p-value is $P(F_{p,n_1+n_2-p-1} \ge F_{\text{obs}})$ with $F_{\text{obs}} = \frac{n_1+n_2-p-1}{(n_1+n_2-2)p} T^2_{\text{obs}}$. For contrasts, $H_0 : \mathbf{C}\boldsymbol{\mu}_1 = \mathbf{C}\boldsymbol{\mu}_2$ ($\mathbf{C}$: $q \times p$):

$$\left(\tfrac{1}{n_1} + \tfrac{1}{n_2}\right)^{-1} \|(\mathbf{C}\mathbf{S}_p\mathbf{C}^\top)^{-1/2}\mathbf{C}(\bar{\mathbf{x}} - \bar{\mathbf{y}})\|_2^2 \sim \frac{q(n_1 + n_2 - 2)}{n_1 + n_2 - q - 1} F_{q,n_1+n_2-q-1}.$$

Theorem 6.6 (Result 6.3): simultaneous intervals with a pooled covariance

Let c_α^2 be as above. With probability $1 - \alpha$,

$$\mathbf{a}^\top(\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2) \pm c_\alpha \sqrt{\mathbf{a}^\top \left(\tfrac{1}{n_1} + \tfrac{1}{n_2}\right)\mathbf{S}_{\text{pooled}}\mathbf{a}}$$

covers $\mathbf{a}^\top(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ for all $\mathbf{a}$. In particular $\mu_{1j} - \mu_{2j}$ is covered by $(\bar{X}_{1j} - \bar{X}_{2j}) \pm c_\alpha \sqrt{(\frac{1}{n_1} + \frac{1}{n_2})[\mathbf{S}_{\text{pooled}}]_{jj}}$, $j = 1, \ldots, p$.

The proof is as for Theorem 5.9, using $\max_{\mathbf{a}} t(\mathbf{a})^2 = T^2$.

2.2 Unequal covariances: large samples

Theorem 6.7 (Result 6.4): simultaneous intervals without a pooled covariance

If $n_1 - p$ and $n_2 - p$ are large, an approximate $100(1-\alpha)\%$ confidence ellipsoid for $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$ is

$$[\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)]^\top \left(\tfrac{1}{n_1}\mathbf{S}_1 + \tfrac{1}{n_2}\mathbf{S}_2\right)^{-1} [\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)] \le \chi^2_p(\alpha),$$

and $\mathbf{a}^\top(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ belongs to $\mathbf{a}^\top(\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2) \pm \sqrt{\chi^2_p(\alpha)}\sqrt{\mathbf{a}^\top(\tfrac{1}{n_1}\mathbf{S}_1 + \tfrac{1}{n_2}\mathbf{S}_2)\mathbf{a}}$ simultaneously for all $\mathbf{a}$.

Sketch. (i) $\boldsymbol{\delta}_n := (\bar{\mathbf{x}}-\bar{\mathbf{y}})-(\boldsymbol{\mu}_1-\boldsymbol{\mu}_2) \sim N_p(\mathbf{0}, \tfrac{1}{n_1}\boldsymbol{\Sigma}_1+\tfrac{1}{n_2}\boldsymbol{\Sigma}_2)$ (exactly under normality, approximately by the CLT), so $\boldsymbol{\delta}_n^\top(\tfrac{1}{n_1}\boldsymbol{\Sigma}_1 + \tfrac{1}{n_2}\boldsymbol{\Sigma}_2)^{-1}\boldsymbol{\delta}_n \sim \chi^2_p$.

(ii) $(n_1-1)\mathbf{S}_1 \sim W_p(n_1-1, \boldsymbol{\Sigma}_1)$, $(n_2-1)\mathbf{S}_2 \sim W_p(n_2-1, \boldsymbol{\Sigma}_2)$, and $\mathbf{S}_i \xrightarrow{p} \boldsymbol{\Sigma}_i$.

(iii) For $n_1 = n_2 = n$: $\mathbf{S}_1 + \mathbf{S}_2 \to \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2$, so $(\mathbf{S}_1 + \mathbf{S}_2)^{-1} \to (\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)^{-1}$ and

$$T^2 = n\boldsymbol{\delta}_n^\top(\mathbf{S}_1 + \mathbf{S}_2)^{-1}\boldsymbol{\delta}_n = n\boldsymbol{\delta}_n^\top(\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)^{-1}\boldsymbol{\delta}_n + n\boldsymbol{\delta}_n^\top\left[(\mathbf{S}_1 + \mathbf{S}_2)^{-1} - (\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)^{-1}\right]\boldsymbol{\delta}_n \xrightarrow{d} \chi^2_p,$$

the second term being $O_p(1)\cdot o_p(1)$. ■

Proof for general n_1, n_2 [Supplement]. Let $n_1, n_2 \to \infty$ (no condition on n_1/n_2), $\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2 \succ 0$, and

$$\mathbf{V}_n = \tfrac{1}{n_1}\boldsymbol{\Sigma}_1 + \tfrac{1}{n_2}\boldsymbol{\Sigma}_2, \qquad \hat{\mathbf{V}}_n = \tfrac{1}{n_1}\mathbf{S}_1 + \tfrac{1}{n_2}\mathbf{S}_2, \qquad T^2 = \boldsymbol{\delta}_n^\top \hat{\mathbf{V}}_n^{-1}\boldsymbol{\delta}_n.$$

Step 1. $\mathbf{Z}_n := \mathbf{V}_n^{-1/2}\boldsymbol{\delta}_n \xrightarrow{d} N_p(\mathbf{0}, \mathbf{I})$ by (i).

Step 2. $\mathbf{M}_n := \mathbf{V}_n^{-1/2}\hat{\mathbf{V}}_n\mathbf{V}_n^{-1/2} \xrightarrow{p} \mathbf{I}$. For a unit $\mathbf{x}$ put $\mathbf{y} = \mathbf{V}_n^{-1/2}\mathbf{x}$; since $\mathbf{V}_n \succeq \boldsymbol{\Sigma}_1/n_1$, $\mathbf{y}^\top\boldsymbol{\Sigma}_1\mathbf{y}/n_1 \le \mathbf{y}^\top\mathbf{V}_n\mathbf{y} = 1$, so

$$\tfrac{1}{n_1}|\mathbf{y}^\top(\mathbf{S}_1 - \boldsymbol{\Sigma}_1)\mathbf{y}| \le \|\boldsymbol{\Sigma}_1^{-1/2}\mathbf{S}_1\boldsymbol{\Sigma}_1^{-1/2} - \mathbf{I}\| \, \tfrac{\mathbf{y}^\top\boldsymbol{\Sigma}_1\mathbf{y}}{n_1} \le \|\boldsymbol{\Sigma}_1^{-1/2}\mathbf{S}_1\boldsymbol{\Sigma}_1^{-1/2} - \mathbf{I}\| \xrightarrow{p} 0$$

by (ii); likewise for sample 2. Hence $\|\mathbf{M}_n - \mathbf{I}\| \le \|\boldsymbol{\Sigma}_1^{-1/2}\mathbf{S}_1\boldsymbol{\Sigma}_1^{-1/2} - \mathbf{I}\| + \|\boldsymbol{\Sigma}_2^{-1/2}\mathbf{S}_2\boldsymbol{\Sigma}_2^{-1/2} - \mathbf{I}\| \xrightarrow{p} 0$.

Step 3. $T^2 = \mathbf{Z}_n^\top\mathbf{M}_n^{-1}\mathbf{Z}_n$ and $\mathbf{M}_n^{-1} \xrightarrow{p} \mathbf{I}$, so $T^2 \xrightarrow{d} \mathbf{Z}^\top\mathbf{Z} \sim \chi^2_p$ (Slutsky).

Step 4. By Cauchy–Schwarz, $\max_{\mathbf{a}}(\mathbf{a}^\top\boldsymbol{\delta}_n)^2/\mathbf{a}^\top\hat{\mathbf{V}}_n\mathbf{a} = T^2$, so all intervals hold together with probability $P(T^2 \le \chi^2_p(\alpha)) \to 1-\alpha$. ■

3 One-Way MANOVA

3.1 Data and assumptions

Random samples from g populations: population ℓ: $\mathbf{X}_{\ell 1}, \mathbf{X}_{\ell 2}, \ldots, \mathbf{X}_{\ell n_\ell}$, $\ell = 1, \ldots, g$. MANOVA (multivariate analysis of variance) asks whether the mean vectors are equal and, if not, which components differ.

1. $\mathbf{X}_{\ell 1}, \ldots, \mathbf{X}_{\ell n_\ell}$ is a random sample of size n_ℓ from a population with mean $\boldsymbol{\mu}_\ell$; samples from different populations are **independent**.
2. All populations have a **common** covariance matrix $\boldsymbol{\Sigma}$.
3. Each population is **multivariate normal** (can be relaxed by the CLT, Theorem 4.11).

3.2 Review: univariate ANOVA

$X_{\ell j} = \mu + \tau_\ell + e_{\ell j}$, $e_{\ell j}$ independent $N(0, \sigma^2)$, with the constraint $\sum_\ell n_\ell \tau_\ell = 0$ and $H_0 : \tau_1 = \cdots = \tau_g = 0$ ($\mu_\ell = \mu + \tau_\ell$, $\tau_\ell = \mu_\ell - \mu$). The decomposition $x_{\ell j} = \bar{x} + (\bar{x}_\ell - \bar{x}) + (x_{\ell j} - \bar{x}_\ell)$ ($\bar{x}$ estimates μ, $\hat{\tau}_\ell = \bar{x}_\ell - \bar{x}$ estimates τ_ℓ, $x_{\ell j} - \bar{x}_\ell$ estimates $e_{\ell j}$) gives

$$\underbrace{\sum_\ell \sum_j (x_{\ell j} - \bar{x})^2}_{SS_{\text{cor}}\ \text{(total)}} = \underbrace{\sum_\ell n_\ell (\bar{x}_\ell - \bar{x})^2}_{SS_{\text{tr}}\ \text{(between)}} + \underbrace{\sum_\ell \sum_j (x_{\ell j} - \bar{x}_\ell)^2}_{SS_{\text{res}}\ \text{(within)}},$$

since the cross term is $\sum_\ell (\bar{x}_\ell - \bar{x}) \sum_j (x_{\ell j} - \bar{x}_\ell) = 0$. Equivalently $SS_{\text{obs}} = SS_{\text{mean}} + SS_{\text{tr}} + SS_{\text{res}}$. Reject H_0 if $F = \dfrac{SS_{\text{tr}}/(g-1)}{SS_{\text{res}}/(\sum n_\ell - g)} > F_{g-1, \sum n_\ell - g}(\alpha)$.

Source	SS	d.f.
Treatments	$SS_{\text{tr}} = \sum_\ell n_\ell (\bar{x}_\ell - \bar{x})^2$	$g - 1$
Residual	$SS_{\text{res}} = \sum_\ell \sum_j (x_{\ell j} - \bar{x}_\ell)^2$	$\sum n_\ell - g$
Total (corrected)	$SS_{\text{cor}} = \sum_\ell \sum_j (x_{\ell j} - \bar{x})^2$	$\sum n_\ell - 1$

3.3 The MANOVA model and the SSP decomposition

Definition 6.8: one-way MANOVA model

$\mathbf{X}_{\ell j} = \boldsymbol{\mu} + \boldsymbol{\tau}_\ell + \mathbf{e}_{\ell j}$, $j = 1, \ldots, n_\ell$, $\ell = 1, \ldots, g$, with $\mathbf{e}_{\ell j}$ independent $N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $\sum_\ell n_\ell \boldsymbol{\tau}_\ell = \mathbf{0}$. (Indeed, $\boldsymbol{\mu} = \mathrm{E}[\bar{\mathbf{X}}] \iff \sum_\ell \sum_j \mathrm{E}(\mathbf{X}_{\ell j}) = n\boldsymbol{\mu} \iff n\boldsymbol{\mu} + \sum n_\ell \boldsymbol{\tau}_\ell = n\boldsymbol{\mu} \iff \sum n_\ell \boldsymbol{\tau}_\ell = \mathbf{0}$.)

Theorem 6.9: SSP decomposition

From $\mathbf{x}_{\ell j} = \bar{\mathbf{x}} + (\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}}) + (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)$ (overall mean $\hat{\boldsymbol{\mu}}$ + estimated treatment effect $\hat{\boldsymbol{\tau}}_\ell$ + residual $\hat{\mathbf{e}}_{\ell j}$),

$$\underbrace{\sum_{\ell=1}^{g} \sum_{j=1}^{n_\ell} (\mathbf{x}_{\ell j} - \bar{\mathbf{x}})(\mathbf{x}_{\ell j} - \bar{\mathbf{x}})^\top}_{\mathbf{T}=\mathbf{B}+\mathbf{W}} = \underbrace{\sum_{\ell=1}^{g} n_\ell (\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}})(\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}})^\top}_{\mathbf{B}\text{: between}} + \underbrace{\sum_{\ell=1}^{g} \sum_{j=1}^{n_\ell} (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)(\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)^\top}_{\mathbf{W}\text{: within}}.$$

$\mathbf{W} = (n_1 - 1)\mathbf{S}_1 + \cdots + (n_g - 1)\mathbf{S}_g$ generalizes $(n_1 + n_2 - 2)\mathbf{S}_{\text{pooled}}$.

Proof. [Supplement] Expanding, the cross terms are $\sum_\ell (\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}}) \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)^\top = \mathbf{0}$. ■

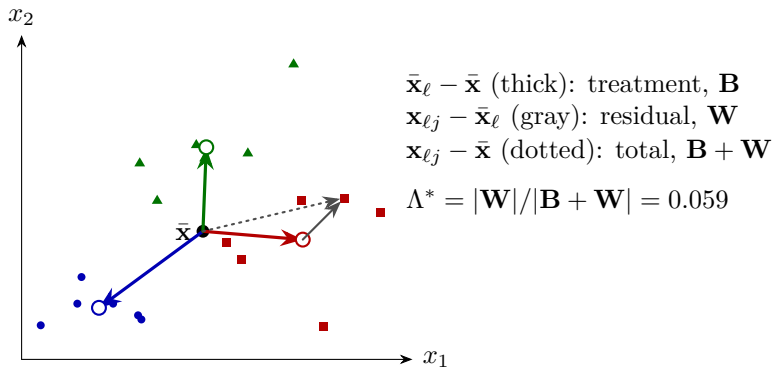


Figure. *One-way MANOVA, $g = 3$, $p = 2$: $\mathbf{x}_{\ell j} - \bar{\mathbf{x}} = (\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}}) + (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)$; groups far apart relative to their spread give a small Λ^*.*

Source	Matrix of SSP	d.f.
Treatment	$\mathbf{B} = \sum_\ell n_\ell(\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}})(\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}})^\top$	$g-1$
Residual (error)	$\mathbf{W} = \sum_\ell \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)(\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)^\top$	$\sum n_\ell - g$
Total	$\mathbf{B}+\mathbf{W} = \sum_\ell \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}})(\mathbf{x}_{\ell j} - \bar{\mathbf{x}})^\top$	$\sum n_\ell - 1$

Remark 6.10: $\mathbf{T} = \mathbf{B} + \mathbf{W}$ as a mixture

Observe $(\mathbf{X}, G)$ with $G \in \{1, \dots, g\}$, $P(G = i) = \pi_i$, $\mathbf{X} \mid G = i \sim N_p(\boldsymbol{\mu}_i, \boldsymbol{\Sigma})$, so $f(\mathbf{x}) = \sum_i f(\mathbf{x} \mid G = i) P(G = i) = \sum_i \phi(\mathbf{x}; \boldsymbol{\mu}_i, \boldsymbol{\Sigma})\pi_i$. Then $\mathrm{E}(\mathbf{X}) = \int \mathbf{x} \sum_i \phi(\mathbf{x}; \boldsymbol{\mu}_i, \boldsymbol{\Sigma})\pi_i \, d\mathbf{x} = \sum_i \pi_i \boldsymbol{\mu}_i = \boldsymbol{\mu}$ and

$$\mathrm{Var}(\mathbf{X}) = \sum_i \pi_i \mathrm{E}[(\mathbf{x} - \boldsymbol{\mu}_i)(\mathbf{x} - \boldsymbol{\mu}_i)^\top \mid G = i] + \sum_i \pi_i (\boldsymbol{\mu}_i - \boldsymbol{\mu})(\boldsymbol{\mu}_i - \boldsymbol{\mu})^\top$$

$$= \boldsymbol{\Sigma} + \sum_i \pi_i (\boldsymbol{\mu}_i - \boldsymbol{\mu})(\boldsymbol{\mu}_i - \boldsymbol{\mu})^\top.$$

The first term is the population version of $\mathbf{W}/n$, the second of $\mathbf{B}/n$.

3.4 Wilks' lambda

Definition 6.11: Wilks' test

Reject $H_0 : \boldsymbol{\tau}_1 = \cdots = \boldsymbol{\tau}_g = \mathbf{0}$ if the ratio of generalized variances

$$\Lambda^* = \frac{|\mathbf{W}|}{|\mathbf{B} + \mathbf{W}|} = \frac{\left|\sum_\ell \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)(\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)^\top\right|}{\left|\sum_\ell \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}})(\mathbf{x}_{\ell j} - \bar{\mathbf{x}})^\top\right|}$$

is **too small**. Proposed by Wilks, it corresponds to the univariate F test and is related to the likelihood ratio criterion.

Theorem 6.12: Wilks' lambda is the likelihood ratio statistic

With $n = \sum n_\ell$, $\Lambda^{2/n} = \Lambda^*$ and $-2 \log \Lambda = n \log \dfrac{|\mathbf{B} + \mathbf{W}|}{|\mathbf{W}|} = n \log \dfrac{|\mathbf{T}|}{|\mathbf{W}|} \dot{\sim} \chi^2_{p(g-1)}$.

Proof. Under H_1: $\hat{\boldsymbol{\mu}}_\ell^{\mathrm{MLE}} = \bar{\mathbf{x}}_\ell$, $\hat{\boldsymbol{\Sigma}}^{\mathrm{MLE}} = \mathbf{S}_p = \dfrac{\sum_\ell (n_\ell - 1)\mathbf{S}_\ell}{n} = \dfrac{\mathbf{W}}{n}$. Under H_0: $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$, $\hat{\boldsymbol{\Sigma}} = \dfrac{\mathbf{T}}{n}$. The maximized likelihoods are $(2\pi)^{-np/2} |\hat{\boldsymbol{\Sigma}}|^{-n/2} e^{-np/2}$, so $\Lambda = (|\mathbf{W}|/|\mathbf{T}|)^{n/2}$ and $-2 \log(L_0/L_1) = n \log \frac{|\mathbf{T}|}{|\mathbf{W}|}$. Reject H_0 if $n \log \frac{|\mathbf{T}|}{|\mathbf{W}|} > \chi^2_{p(g-1),\alpha}$. ■

Theorem 6.13: distribution of Wilks' lambda for multivariate normal data

No. of variables	No. of groups	Sampling distribution
$p = 1$	$g \geq 2$	$\left(\frac{\sum n_\ell - g}{g-1}\right)\left(\frac{1-\Lambda^*}{\Lambda^*}\right) \sim F_{g-1,\sum n_\ell - g}$
$p = 2$	$g \geq 2$	$\left(\frac{\sum n_\ell - g - 1}{g-1}\right)\left(\frac{1-\sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\right) \sim F_{2(g-1),2(\sum n_\ell - g - 1)}$
$p \geq 1$	$g = 2$	$\left(\frac{\sum n_\ell - p - 1}{p}\right)\left(\frac{1-\Lambda^*}{\Lambda^*}\right) \sim F_{p,\sum n_\ell - p - 1}$
$p \geq 1$	$g = 3$	$\left(\frac{\sum n_\ell - p - 2}{p}\right)\left(\frac{1-\sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\right) \sim F_{2p,2(\sum n_\ell - p - 2)}$

The proof is omitted.

Derivation: the $g = 2$ row — relation to the two-sample T^2

[Supplement] For $g = 2$, $\bar{\mathbf{x}} = (n_1\bar{\mathbf{x}}_1 + n_2\bar{\mathbf{x}}_2)/n$, so $\bar{\mathbf{x}}_1 - \bar{\mathbf{x}} = \frac{n_2}{n}\mathbf{d}$ and $\bar{\mathbf{x}}_2 - \bar{\mathbf{x}} = -\frac{n_1}{n}\mathbf{d}$ with $\mathbf{d} = \bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2$, and

$$\mathbf{B} = n_1\frac{n_2^2}{n^2}\mathbf{d}\mathbf{d}^\top + n_2\frac{n_1^2}{n^2}\mathbf{d}\mathbf{d}^\top = \frac{n_1 n_2}{n}\mathbf{d}\mathbf{d}^\top.$$

By the determinant lemma $|\mathbf{W}+\mathbf{B}| = |\mathbf{W}|(1+\frac{n_1 n_2}{n}\mathbf{d}^\top\mathbf{W}^{-1}\mathbf{d})$, and $\mathbf{W} = (n-2)\mathbf{S}_{\text{pooled}}$ gives $\frac{n_1 n_2}{n}\mathbf{d}^\top\mathbf{W}^{-1}\mathbf{d} = \frac{T^2}{n-2}$. Hence

$$\Lambda^* = \left(1 + \frac{T^2}{n-2}\right)^{-1}, \qquad \frac{1-\Lambda^*}{\Lambda^*} = \frac{T^2}{n-2},$$

and Theorem 6.5 ($\frac{n-p-1}{(n-2)p}T^2 \sim F_{p,n-p-1}$) gives the third row. Wilks' test with $g = 2$ is the two-sample T^2 test.

Theorem 6.14: Bartlett's test for Wilks' lambda

If H_0 is true and $\sum n_\ell = n$ is large,

$$-\left(n - 1 - \frac{p+g}{2}\right)\ln\Lambda^* = -\left(n - 1 - \frac{p+g}{2}\right)\ln\left(\frac{|\mathbf{W}|}{|\mathbf{B}+\mathbf{W}|}\right) \;\dot{\sim}\; \chi^2_{p(g-1)},$$

so reject H_0 at level α if the left side exceeds $\chi^2_{p(g-1)}(\alpha)$.

[Supplement] The degrees of freedom $p(g-1)$ are the difference in the number of mean parameters (g mean vectors versus one); the factor $n - 1 - (p+g)/2$ improves the chi-square approximation.

Remark 6.15: other test statistics

With the eigenvalues $\eta_1 \geq \cdots \geq \eta_s$ of $\mathbf{W}^{-1}\mathbf{B}$ ($s = \min(p, g-1)$),

$$\frac{|\mathbf{T}|}{|\mathbf{W}|} = |\mathbf{I}_p + \mathbf{W}^{-1/2}\mathbf{B}\mathbf{W}^{-1/2}| = \prod_i(1+\eta_i) = \frac{1}{\Lambda^*}, \qquad \text{Lawley–Hotelling } \operatorname{tr}(\mathbf{W}^{-1}\mathbf{B}) = \sum\eta_i,$$

$$\text{Pillai } \operatorname{tr}\{\mathbf{B}(\mathbf{B}+\mathbf{W})^{-1}\} = \sum\frac{\eta_i}{1+\eta_i}, \qquad \text{Roy } \eta_1 = \|\mathbf{W}^{-1}\mathbf{B}\|_2.$$

[Supplement] All summarize "$\mathbf{B}$ is large relative to $\mathbf{W}$". The η_i are also the eigenvalues of Fisher's discriminant analysis (Ch. 10).

Example 6.16 (Example 6.10): Wisconsin nursing home data

Costs per patient-day: X_1 nursing labour, X_2 dietary labour, X_3 plant operation and maintenance labour, X_4 housekeeping and laundry labour. Ownership: private ($n_1 = 271$, $\bar{\mathbf{x}}_1^\top = (2.066, .480, .082, .360)$), non-profit ($n_2 = 138$, $\bar{\mathbf{x}}_2^\top = (2.167, .596, .124, .418)$), government ($n_3 = 107$, $\bar{\mathbf{x}}_3^\top = (2.273, .521, .125, .383)$), $\sum n_\ell = 516$. The $\mathbf{S}_\ell$ look compatible and are pooled:

$$\mathbf{W} = \begin{pmatrix} 182.962 & & & \\ 4.408 & 8.200 & & \\ 1.695 & 0.633 & 1.484 & \\ 9.581 & 2.428 & 0.934 & 6.538 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 3.475 & & & \\ 1.111 & 1.225 & & \\ 0.821 & 0.453 & 0.235 & \\ 0.584 & 0.610 & 0.230 & 0.304 \end{pmatrix}$$

(symmetric; lower triangles shown), $\bar{\mathbf{x}} = (2.136, 0.519, 0.102, 0.380)^\top$. With $\Lambda^* = 0.7714$, $p = 4$, $g = 3$:

$$\Big(\frac{\sum n_\ell - p - 2}{p}\Big)\Big(\frac{1 - \sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\Big) = \Big(\frac{516 - 4 - 2}{4}\Big)\Big(\frac{1 - \sqrt{.7714}}{\sqrt{.7714}}\Big) = 17.67 > F_{8,1020}(.01) \approx 2.51.$$

Reject H_0 at the 1% level: average costs differ by type of ownership. ([Supplement] Bartlett: $-(516 - 1 - 3.5)\ln 0.7714 = 132.8 > \chi^2_8(.01) = 20.09$.)

4 Simultaneous Intervals for Treatment Effects

Let τ_{kj} be the jth component of $\boldsymbol{\tau}_k$. Since $\hat{\boldsymbol{\tau}}_k = \bar{\mathbf{X}}_k - \bar{\mathbf{X}}$, $\hat{\tau}_{kj} - \hat{\tau}_{\ell j} = \bar{X}_{kj} - \bar{X}_{\ell j}$ and a two-sample t-based interval is valid with an appropriately corrected α:

$$\mathrm{Var}(\hat{\tau}_{kj} - \hat{\tau}_{\ell j}) = \Big(\frac{1}{n_k} + \frac{1}{n_\ell}\Big)\sigma_{jj}, \qquad \widehat{\mathrm{Var}}(\bar{X}_{kj} - \bar{X}_{\ell j}) = \Big(\frac{1}{n_k} + \frac{1}{n_\ell}\Big)\frac{\mathbf{W}_{jj}}{n - g}.$$

Apportioning the error rate: for p variables and $g(g-1)/2$ pairwise differences there are $m = p\,g(g-1)/2$ statements, so each interval uses $t_{n-g}(\alpha/2m)$.

Theorem 6.17 (Result 6.5): simultaneous intervals for differences in treatment effects

Let $n = \sum_k n_k$. With confidence at least $1 - \alpha$,

$$\tau_{kj} - \tau_{\ell j} \;\text{ belongs to }\; \bar{x}_{kj} - \bar{x}_{\ell j} \pm t_{n-g}\Big(\frac{\alpha}{pg(g-1)}\Big)\sqrt{\frac{\mathbf{W}_{jj}}{n-g}\Big(\frac{1}{n_k} + \frac{1}{n_\ell}\Big)}$$

for all components $j = 1, \ldots, p$ and all differences $\ell < k = 1, \ldots, g$.

Proof. [Supplement] $W_{jj}/\sigma_{jj} \sim \chi^2_{n-g}$ independent of the group means, so each studentized difference is t_{n-g} and each interval has coverage $1 - \alpha/m$; Bonferroni (Theorem 5.11) over the m statements gives at least $1 - \alpha$. ■

Example 6.18 (Example 6.11): nursing home cost comparison

Compare plant operation and maintenance (X_3) between private and government homes, i.e. estimate $\tau_{13} - \tau_{33}$: $\hat{\boldsymbol{\tau}}_1 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}) = (-.070, -.039, -.020, -.020)^\top$, $\hat{\boldsymbol{\tau}}_3 = (\bar{\mathbf{x}}_3 - \bar{\mathbf{x}}) = (.137, .002, .023, .003)^\top$, so $\hat{\tau}_{13} - \hat{\tau}_{33} = -.020 - .023 = -.043$. With $n = 271 + 138 + 107 = 516$: $\sqrt{(\frac{1}{271} + \frac{1}{107})\frac{1.484}{516-3}} = 0.00614$; for $p = 4$, $g = 3$ the 95% interval uses $t_{513}(.05/4(3)2) =$

$t_{513}(.00208) \approx 2.87$. Hence $\tau_{13}-\tau_{33} \in -0.043 \pm 2.87 \times 0.00614 = (-0.061, -0.025)$: government homes have higher plant operation costs.

5 Two-Way MANOVA

5.1 Review: univariate two-way ANOVA

Factor 1 with g levels, factor 2 with b levels, n independent observations at each of the gb combinations; $X_{\ell kr}$ is the rth observation at level ℓ of factor 1 and level k of factor 2:

	$k=1$	$k=2$	$\cdots$	$k=b$	
$\ell=1$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\cdots$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\bar{\mathbf{x}}_{\ell\cdot}$
$\ell=2$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\cdots$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\bar{\mathbf{x}}_{\ell\cdot}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	
$\ell=g$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\cdots$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\bar{\mathbf{x}}_{\ell\cdot}$
	$\bar{\mathbf{x}}_{\cdot k}$	$\bar{\mathbf{x}}_{\cdot k}$		$\bar{\mathbf{x}}_{\cdot k}$	$\bar{\mathbf{x}}$

factor 1 margin (mean of bn obs.)

factor 2 margin (mean of gn obs.)

Figure. *Two-way layout: $g \times b$ cells with n replicates each; cell, row, column and grand means.*

$$X_{\ell kr} = \underbrace{\mu}_{\text{overall mean}} + \underbrace{\tau_\ell}_{\text{factor 1 effect}} + \underbrace{\beta_k}_{\text{factor 2 effect}} + \underbrace{\gamma_{\ell k}}_{\text{interaction}} + \underbrace{e_{\ell kr}}_{\text{normal error}},$$

$\sum_\ell \tau_\ell = \sum_k \beta_k = \sum_\ell \gamma_{\ell k} = \sum_k \gamma_{\ell k} = 0$, and $SS_{\text{cor}} = SS_{\text{fac1}} + SS_{\text{fac2}} + SS_{\text{int}} + SS_{\text{res}}$:

Source	Sum of squares	d.f.
Factor 1	$SS_{\text{fac1}} = \sum_\ell bn(\bar{x}_{\ell\cdot} - \bar{x})^2$	$g-1$
Factor 2	$SS_{\text{fac2}} = \sum_k gn(\bar{x}_{\cdot k} - \bar{x})^2$	$b-1$
Interaction	$SS_{\text{int}} = \sum_\ell \sum_k n(\bar{x}_{\ell k} - \bar{x}_{\ell\cdot} - \bar{x}_{\cdot k} + \bar{x})^2$	$(g-1)(b-1)$
Residual	$SS_{\text{res}} = \sum_\ell \sum_k \sum_r (x_{\ell kr} - \bar{x}_{\ell k})^2$	$gb(n-1)$
Total	$SS_{\text{cor}} = \sum\sum\sum (x_{\ell kr} - \bar{x})^2$	$gbn-1$

F-ratios: $F_{\text{fac1}} = \frac{SS_{\text{fac1}}/(g-1)}{SS_{\text{res}}/(gb(n-1))}$, $F_{\text{fac2}} = \frac{SS_{\text{fac2}}/(b-1)}{SS_{\text{res}}/(gb(n-1))}$, $F_{\text{int}} = \frac{SS_{\text{int}}/((g-1)(b-1))}{SS_{\text{res}}/(gb(n-1))}$.

5.2 Multivariate two-way fixed-effects model with interaction

Definition 6.19: two-way MANOVA

$\mathbf{X}_{\ell kr} = \boldsymbol{\mu} + \boldsymbol{\tau}_\ell + \boldsymbol{\beta}_k + \boldsymbol{\gamma}_{\ell k} + \mathbf{e}_{\ell kr}$ with $\mathbf{e}_{\ell kr}$ independent $N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $\sum_\ell \boldsymbol{\tau}_\ell = \sum_k \boldsymbol{\beta}_k = \sum_\ell \boldsymbol{\gamma}_{\ell k} = \sum_k \boldsymbol{\gamma}_{\ell k} = \mathbf{0}$; all vectors are $p \times 1$.

The SSP decomposition $\text{SSP}_{\text{cor}} = \text{SSP}_{\text{fac1}} + \text{SSP}_{\text{fac2}} + \text{SSP}_{\text{int}} + \text{SSP}_{\text{res}}$ replaces each $(\cdot)^2$ by $(\cdot)(\cdot)^\top$, e.g. $\text{SSP}_{\text{int}} = \sum_\ell \sum_k n(\bar{\mathbf{x}}_{\ell k} - \bar{\mathbf{x}}_{\ell\cdot} - \bar{\mathbf{x}}_{\cdot k} + \bar{\mathbf{x}})(\bar{\mathbf{x}}_{\ell k} - \bar{\mathbf{x}}_{\ell\cdot} - \bar{\mathbf{x}}_{\cdot k} + \bar{\mathbf{x}})^\top$, with the same degrees of freedom.

Theorem 6.20: two-way MANOVA tests (Bartlett-corrected)

Interaction, $H_0 : \boldsymbol{\gamma}_{11} = \boldsymbol{\gamma}_{12} = \cdots = \boldsymbol{\gamma}_{gb} = \mathbf{0}$: with $\Lambda^* = \dfrac{|\mathrm{SSP}_{\mathrm{res}}|}{|\mathrm{SSP}_{\mathrm{int}} + \mathrm{SSP}_{\mathrm{res}}|}$, reject if

$$-\Big[gb(n-1) - \frac{p+1-(g-1)(b-1)}{2}\Big] \ln \Lambda^* > \chi^2_{(g-1)(b-1)p}(\alpha).$$

Main effects of factor 1, $H_0 : \boldsymbol{\tau}_1 = \cdots = \boldsymbol{\tau}_g = \mathbf{0}$: with $\Lambda^* = \dfrac{|\mathrm{SSP}_{\mathrm{res}}|}{|\mathrm{SSP}_{\mathrm{fac1}} + \mathrm{SSP}_{\mathrm{res}}|}$, reject if

$$-\Big[gb(n-1) - \frac{p+1-(g-1)}{2}\Big] \ln \Lambda^* > \chi^2_{(g-1)p}(\alpha).$$

[Supplement] Factor 2 is the same with $g-1$ replaced by $b-1$. Test the interaction first; if present, interpret main effects with care.

Theorem 6.21: simultaneous intervals in two-way MANOVA

With $\nu = gb(n-1)$ and E_{jj} the jth diagonal element of $\mathbf{E} = \mathrm{SSP}_{\mathrm{res}}$,

$$\tau_{\ell j} - \tau_{mj} \quad \text{belongs to} \quad (\bar{x}_{\ell\cdot j} - \bar{x}_{m\cdot j}) \pm t_\nu\Big(\frac{\alpha}{pg(g-1)}\Big)\sqrt{\frac{E_{jj}}{\nu}\cdot\frac{2}{bn}}.$$

Proof. [Supplement] $\bar{X}_{\ell\cdot j} - \bar{X}_{m\cdot j} \sim N(\tau_{\ell j} - \tau_{mj}, \frac{2}{bn}\sigma_{jj})$ and $E_{jj}/\sigma_{jj} \sim \chi^2_\nu$ independently; Bonferroni over the $m = p\,g(g-1)/2$ statements. ■